ADANI UNIVERSITY • FACULTY OF ENGINEERING & TECHNOLOGY

Indian Railways Train Delay Prediction

Network Punctuality Analytics & Proactive Dispatch Forecasting

Student Project Team & Roles

• **Dhruv:** Team Lead — Problem Definition, Dataset Scope & Predictor Demo
• **Darshi:** Team Member — EDA Insights, Preprocessing Pipeline & Feature Engineering
• **Kirtan:** Team Member — Algorithm Benchmarks, LightGBM Tuning & Evaluation

Course: ECSCI24302 Machine Learning Essentials • PBL Project Defense
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Intelligent Transportation Analytics Platform

• End-to-end Machine Learning pipeline applied to 10,000 operational journeys.
• Zero data leakage with rigorous NumPy IQR Winsorization and 5 domain features.
• Champion Tuned LightGBM model: **85.85% Accuracy** and **92.10% Recall**.
• Tested 13-module Streamlit operations platform with real-time operational presets.

The Problem: Reactive vs. Proactive Dispatching

Why traditional delay management fails and how proactive ML transforms operations.

Traditional Reactive Dispatching (Firefighting)	Our Proactive Machine Learning Solution
• **Late Awareness:** Controllers only discover delays after trains halt at outer signals. • **Cascading Gridlock:** Missing junction slots paralyzes platforms and connecting trains. • **Passenger Uncertainty:** 23M daily passengers face sudden unscheduled delays. • **Rake Mismanagement:** Delayed incoming trains disrupt turnaround rakes.	• **Pre-Departure Risk Forecasting:** Estimates delay risk probability prior to dispatch. • **Official 15-Minute Threshold:** Adheres strictly to Ministry of Railways punctuality standards. • **3-Tier Action Advisories:** Green (Normal), Amber (Advisory), and Red Alert tiers. • **High 92.10% Recall:** Catches over 9 out of 10 delayed trains before arrival.

END-TO-END SYSTEM DESIGN

The Solution: Intelligent Delay Risk Architecture

A streamlined pipeline from pre-departure data to operational dispatch intelligence.

1. Pre-Departure Inputs	2. Zero-Leakage ML Pipeline	3. Actionable Output
• Scheduled distance & hours	• Constant features purged	• Calibrated delay probability (0–100%)
• Average kinematic speed	• Target leakage quarantined	• **GREEN (<40%):** On-Time Expected
• Route historical on-time %	• NumPy IQR Winsorization	• **AMBER (40–65%):** Caution Buffer
• Zonal track congestion index	• 5 domain engineered features	• **RED ALERT (>65%):** Priority Clear
• Winter fog & seasonal severity	• Cyclical time trigonometry	• Dispatch advisory for DOM
• Rolling stock & coach age	• 8,111 sparse one-hot features	• Standby turnaround relief rake
• Late incoming rake status	• Tuned LightGBM classifier	

DATA ACQUISITION & INTEGRITY

Dataset Overview & Data Quality Assurance

Rigorous dataset exploration, zero-leakage quarantine, and balanced stratified validation.

Data Cleaning & Leakage Quarantine Protocol

- **Full Dataset Scope:** Sourced from Kaggle Indian Railways repository (1.5M rows, 45 attributes).
- **Experimental Sample:** 10,000 stratified journeys (80% train, 20% validation).
- **Zero-Variance Dropped:** Column is_overloaded (constant 0) purged.
- **Target Leakage Quarantined:** Features delay_minutes and primary_delay_cause strictly isolated.

Class Balance & Imbalance Characteristics

- **Empirical Split:** 72.16% delayed vs 27.84% on-time journeys.
- **Operational Cause:** Reflects high traffic density across mixed passenger-freight corridors.
- **Evaluation Strategy:** F1-score and Recall prioritized over simple raw accuracy.
- **Zero Leakage:** All scaling and IQR bounds fitted strictly on training data.

EXPLORATORY DATA ANALYSIS

Key Operational Drivers of Train Delays

Empirical data proves non-linear relationships across speed, weather, and turnaround.

1. Kinematic Speed Threshold

• Average speed <50 km/h exhibits an **88.4% delay probability**.
• Speed >75 km/h drops delays below **15%**.
• Slow trains get repeatedly looped and overtaken by express traffic.

2. Zonal Congestion & Fog

• Northern (NR), North Central (NCR), and Eastern (ER) zones suffer >85% congestion.
• Dense winter fog in Indo-Gangetic plains enforces automatic speed restrictions.

3. Cascading Rake Turnaround

• When a shared incoming rake arrives late (late_incoming_rake = 1), outbound delay probability surges to **91.2%**.
• Delays cascade across schedules.

PREPROCESSING & FEATURE ENGINEERING

4-Stage Preprocessing & Feature Engineering

Preserving extreme train journeys with NumPy IQR Winsorization and 5 domain features.

Stage 1 & 2: Cleaning & NumPy IQR Winsorization

• Constant is_overloaded dropped; identifiers removed.
• Rather than deleting extreme cross-country journeys (e.g. 4,000-km Vivek Express), applied robust IQR Winsorization:
- distance_km capped at **1,466 km**.
- scheduled_travel_hours capped at **24.9 hours**.
• Zero data rows deleted; 100% sample preservation.

Stage 3 & 4: 5 Domain Features & Encoding

• Engineered 5 high-impact domain variables:
1. avg_speed_kmh: Distance / Travel Hours.
2. network_vulnerability: Congestion weighted by HDN.
3. equipment_age_index: Average loco & coach age.
4. rake_turnaround_risk: Shared rake × late arrival.
5. Cyclical Time Encodings: $\sin / \cos$ of departure hour.
• One-hot encoded into 8,111 aligned sparse columns.

MODEL DEVELOPMENT METHODOLOGY

Algorithms Tested: From Baseline to Gradient Boosting

Evaluating diverse supervised learning paradigms against a dummy baseline.

Algorithm Family	Mathematical Mechanism	Validation Benchmark
Baseline (Dummy)	Majority class predictor (always predicts delayed).	Acc: 72.15% | Recall: 100% | ROC-AUC: **0.5000**
Decision Tree	Single recursive binary splitting hierarchy (CART).	Acc: 81.30% | Prec: 85.85% | ROC-AUC: 0.8593
Random Forest	Ensemble bagging of 150 de-correlated decision trees.	Acc: 78.70% | Recall: 98.54% | ROC-AUC: 0.8854
Logistic Regression	L2 ridge regularized linear sigmoid probability boundary.	Acc: 85.70% | Prec: 89.01% | ROC-AUC: **0.9229**
Tuned LightGBM	Leaf-wise gradient boosted trees with histogram binning.	Acc: **85.85%** | Rec: **92.10%** | F1: **0.9038**

MODEL BENCHMARKS & EVALUATION

Validation Performance Comparison (n = 2,000)

LightGBM emerges as the definitive champion with 85.85% accuracy and 0.9038 F1-score.

Model Performance Benchmark Table

- **Dummy Baseline:** Acc: 72.15% | Recall: 100% | F1: 0.8382 | AUC: 0.5000
- **Decision Tree:** Acc: 81.30% | Prec: 85.85% | F1: 0.8725 | AUC: 0.8593
- **Random Forest:** Acc: 78.70% | Prec: 77.83% | F1: 0.8697 | AUC: 0.8854
- **Logistic Regression:** Acc: 85.70% | Prec: 89.01% | F1: 0.9023 | AUC: 0.9229
- **Tuned LightGBM (Champion): 85.85% Acc | 88.72% Prec | 92.10% Rec | 0.9038 F1 | 0.9121 AUC**

Selection Justification & Hyperparameters

- **Why LightGBM Won:** Highest accuracy (85.85%), highest recall (92.10%), and highest F1-score (0.9038).
- **Imbalance Metric:** F1-score is the gold standard on imbalanced data as it balances precision and recall.
- **Optimal Parameters:** num_leaves: 31, learning_rate: 0.05, n_estimators: 200, subsample: 0.8.
- **Inference Latency:** <25ms per train for live dispatching.

DIAGNOSTICS & ERROR ANALYSIS

Confusion Matrix & Error Diagnostics (n = 2,000)

High 92.10% Recall ensures delayed trains are caught early to prevent junction gridlock.

LightGBM Confusion Matrix Breakdown

- **True Positives (1,329 Trains):** Delayed trains correctly flagged in advance.
- **True Negatives (388 Trains):** On-time trains correctly cleared for normal schedule.
- **False Positives (169 Trains / 8.45%):** On-time trains flagged as delayed (triggers precautionary monitoring).
- **False Negatives (114 Trains / 5.70%):** Delayed trains flagged as on-time (missed delays).

Operational Risk & Recall Justification

- **Cost Asymmetry:** In railway operations, a **False Negative is far more hazardous** than a False Positive.
- **Missing a Delay:** Blocks station platforms and triggers cascading network ripple.
- **Precautionary Alert:** False Positive merely triggers cautionary buffer checks.
- **High 92.10% Recall:** Successfully catches over 92 out of 100 delayed trains.

SYSTEM DEMONSTRATION

Live Web Platform & Operational Presets

Real-time decision support tested across premier corridors and bottleneck weather scenarios.

Preset 1: Premier Vande Bharat Express

- High-speed corridor (130 km/h), Northern Railway.
- Double electrified track, clear summer weather.
- Pristine modern LHB rake, 0 min turnaround delay.
- Historical route on-time: 88.5%.

Result: 3.3% Delay Probability -> GREEN: ON-TIME EXPECTED
Advises standard priority dispatch corridor.

Preset 2: Howrah–Amritsar Mail in Heavy Monsoon

- Slow schedule (38 km/h), Eastern Railway.
- Single track non-electrified line, 24 scheduled halts.
- Heavy monsoon rainfall severity (0.90), shared rake.
- Late incoming turnaround rake (1).

Result: 98.7% Delay Probability -> RED ALERT: HIGH DELAY RISK
Urgent advisory: alert DOM, clear turnaround buffers.

ANALYSIS & FUTURE WORK

Key Findings, Limitations & Future Roadmap

Honest engineering analysis of model capabilities, edge cases, and expansion paths.

What We Learned

- Kinematics & rake turnaround explain >85% of predictable variance.
- Pre-departure scheduling provides robust early signals before dispatch.
- Feature engineering added +6.4% F1-score improvement.

Project Limitations

- Operates on pre-departure timetables; does not ingest mid-journey GPS.
- Sudden freight interference causes unpredictable False Negatives.
- 10,000-row sample for local and cloud memory constraints.

Future Roadmap

- Live National Train Enquiry System (NTES) GPS feed integration.
- IoT track-vibration sensors for automated speed restriction monitoring.
- Multi-agent dispatch coordination across divisions.

PROJECT DEFENSE CONCLUSION

Summary of Outcomes & Open Q&A; Defense

Engineering a production-ready machine learning platform for Indian Railways operations.

Project Milestones Achieved

✓ **End-to-End Pipeline:** From raw Kaggle dataset to verified LightGBM inference engine.
✓ **Benchmark Performance:** 85.85% Accuracy, 92.10% Recall, 0.9038 F1-Score, and 0.9121 ROC-AUC.
✓ **Academic Rigor:** Zero data leakage, NumPy IQR Winsorization, and 100% PBL compliance.
✓ **Tested 13-Module Suite:** Real-time inference, batch fleet audit, and threshold explorer.

Thank You • Open for Questions

Student Team:
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• **Darshi** (Team Member)
• **Kirtan** (Team Member)

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We welcome your questions and evaluation.